

Observation Priority v2

Ranks the basin by where an additional observation may carry high scientific value - and enforces, in code, the claim it is not allowed to make.

PORT	PAGE FILE	FORMULA	CELLS RANKED
8516	app/phase2/priority_page.py	$\text{sqrt}(\text{sigma} \times \text{EKE})$	8,790

Ocean observation is expensive. If you could deploy one more instrument, where should it go? This page offers an answer - and is unusually strict about the difference between offering an answer and claiming authority.

1. What this feature is, in plain words

Two things make a location worth measuring. The first is that **the model is uncertain** there, because a measurement where the model is already confident teaches it little. The second is that **the water is energetic** there - full of eddies and variability - because that is where a single measurement carries the most information about a rapidly changing state.

So the score is the geometric mean of those two: the model's own uncertainty, and eddy kinetic energy. Both are normalised to a common scale first, because uncertainty is in degrees Celsius and energy is in metres squared per second squared - a raw product would simply rank on whichever number happens to be larger.

Version 1 used float sparsity instead of energy, and it had a defect that could not be fixed from inside it. Sparsity asks 'where are there no floats?' - and the places floats are most absent are frequently the places floats **cannot go**. Version 1 ranked the Persian Gulf, about 20 metres deep, at the very top. Energy asks a question about the water rather than about the observing network, and cannot be maximised by a place no instrument can reach.

And here is the discipline a jury should notice. Our own literature review marks observation-priority as **already done** - it is a simplified version of a formally optimised research area. So the sanctioned wording lives **inside the code** and is rendered verbatim, and a test searches both the module and the page for any phrasing claiming the model tells anyone where to deploy floats, and fails unless it is negated. *No numerical test can catch a rhetorical error, so the rhetoric is tested too.*

THE EVERYDAY PICTURE

A doctor deciding where to place one more sensor puts it where they are least certain and where the patient is most variable - not where sensors happen to be scarce, because some places are scarce for good reasons.

2. What you actually see on the screen

A ranked map with its own claim printed on it:

- A basin map ranked by priority score, with the top candidate cells highlighted.
- The two contributing factors shown separately, so a viewer can see which one is driving any given cell.
- The sanctioned claim and the explicitly disallowed claim, rendered verbatim from the code rather than typed into the page.
- A switch that turns the too-shallow guard **off**, labelled as reproducing version 1's failure mode - so the improvement is demonstrable rather than asserted.

- The number of days used in the mean-flow window, reported rather than assumed.

3. How to run it, step by step

RUN THIS

```
PYTHONPATH=src python -m streamlit run app/phase2/priority_page.py --server.port 8516
```

- 1 Run the command and open **http://localhost:8516**.
- 2 Pick a date in the southwest monsoon for the clearest picture.
- 3 Read the claim box at the top out loud. It says *regions where additional observations may provide high scientific value* - and it does **not** say the model tells anyone where to deploy anything.
- 4 Look at the top candidates. They cluster at 7 N, 50-54 E - the Somali Current and its great eddy, the most energetic field in the basin during the southwest monsoon, and a place where the model is also uncertain.
- 5 Toggle the shallow-water guard off and let the audience watch version 1's failure mode reappear.
- 6 Finish with the honest measurement below - the guard's mechanism matters, but on this date it rescued nothing, and we say so.

4. Why the problem statement needs this

The problem statement's sponsor operates an observing network. A layer that suggests where additional observations would be most informative is directly relevant to how that network is planned - and it closes the loop from reconstruction back to observation.

It also demonstrates something a jury rarely sees: a team that found its own idea was already published, said so in writing, kept the feature because it is useful, and dropped the novelty claim. That is recorded in the project's novelty matrix, which corrected its own earlier assessment.

PS requirement	How this feature answers it
Products beyond raw reconstruction	Turns the uncertainty field into an observation-planning layer.
Honest positioning	The claim is bounded in code and enforced by a test, not left to a presenter's judgement.
Uses the model's own uncertainty	Consumes the calibrated per-depth sigma from Feature 13 rather than a proxy.

5. The measured numbers behind it

Four checks, because a plausible-looking map is the easiest thing in this project to get wrong [VERIFIED]:

Check	Result	What it rules out
Is this just a picture of the fastest currents?	Top-decile energy against top-decile mean speed overlap 0.305	The map is not ranking the famous, well-understood strong currents. Energy must come from the current's <i>anomaly</i> , not its total.
Does a steady jet score zero?	A perfectly steady 1.5 m/s jet scores under 1e-20	The cleanest test of the whole idea: a strong but unvarying current carries no information and must rank at the bottom.
Is the anomaly really an anomaly?	Maximum time-mean of the anomaly: 5.7e-16 m/s	Numerically zero - the mean flow really has been removed.
Did the shallow-water guard actually rescue the map?	It excludes 3,042 too-shallow cells, and none of them were in the top 200 either way	The bug's mechanism is gone - the guard did not save this particular map. Both statements are on the page.

- The mean flow is removed over a **rolling 30-day window**, not over the whole record. This basin reverses its circulation seasonally, so subtracting a full-record mean would leave the monsoon reversal inside the anomaly and the 'energy' would simply be the seasonal cycle, large almost everywhere.
- The two factors are combined by the **same** geometric-mean method version 1 used, deliberately - so a version 1 versus version 2 comparison measures the factor swap and not a change of scaling. A test asserts the two rank identically when version 1 is given a neutral third factor.
- **8,790 cells ranked.** Top candidates cluster at 7 N, 50-54 E - the Somali Current and Great Whirl region.

6. The impact - and who it is for

Who	What changes for them
Observation-network planners	A data-driven starting point for where an additional deployment might be most informative.
Research cruise planning	Suggests where ship time would return the most information.
The modelling team	Closes the loop - the model's own weaknesses point at what would most improve it.
The jury	A worked example of a team bounding its own claim in code because prose is not enough.

WHY SOCIETY GAINS FROM THIS

Ocean observation is publicly funded and severely limited by cost. Any tool that helps place a limited number of instruments where they will teach the most makes that public money go further - provided it is offered as a suggestion to an expert rather than as an instruction, which is exactly the line this feature enforces.

7. Say this to the jury

If you could deploy one more instrument, where should it go? We rank the basin by two things: where the model is uncertain, and where the water is energetic. The top candidates cluster at 7 north, 50 to 54 east - the Somali Current and the Great Whirl. But I want you to see how carefully this is claimed. Our own literature review says observation-priority is already done and done better, so the sanctioned wording lives in the code and is rendered verbatim, and a test greps both the module and the page for any phrasing saying the model tells anyone where to deploy floats. We test the rhetoric, because no numerical test can catch a rhetorical error.

Testing the wording is a genuinely unusual practice - point it out explicitly.

8. Questions the jury will ask

Q. Is this not just showing you the Somali Current?

A. We tested exactly that objection. The overlap between the top decile of our energy field and the top decile of mean current speed is only 0.305, and a perfectly steady 1.5 m/s jet scores under 1e-20 in our metric. We rank variability, not speed - a strong but unvarying current teaches you nothing.

Q. Are you claiming novelty for this?

A. No, explicitly. Our novelty matrix marks observation-priority as already done, and names the formally-optimised prior work. Ours is a simple heuristic with no cost model, no float drift physics and no budget constraint. We kept it because it is useful and we dropped the novelty claim because it is not ours.

Q. What was wrong with the first version?

A. It used float sparsity - distance to the nearest float - and the places floats are most absent are often the places floats cannot go. It ranked the Persian Gulf, about 20 metres deep, at the top. Energy asks about the water rather than

about the observing network, so the failure mode is structurally gone. And the page has a switch to turn the guard off so you can watch the old failure reappear.

9. The honest limit - say it before they find it

A HEURISTIC, AND IT SAYS SO

This is a simplified version of an established research area with dedicated formal methods - optimising float deployment to minimise mapping uncertainty, differentiable sensor placement, and more. Ours has no cost model, no float drift physics and no budget constraint, and one of its two inputs - the model's sigma - was itself measured to be mildly overconfident. It is a suggestion to an expert, framed as one, and never as an instruction to an agency.

IN ONE LINE

Ranks where another observation would teach the model most - and enforces the boundary of that claim with a test on its own wording.

8,790 CELLS RANKED	0.305 OVERLAP WITH MEAN SPEED - IT IS NOT JUST THE CURRENT	under 1e-20 SCORE FOR A STEADY JET, AS IT MUST BE
--	--	---

Feature 16 of 17. Next: Feature 17.